AI-based Eye Movement and Pupil Dilation Analyzer for Cognitive Load Detection

1. Problem Statement

In modern digital environments (e-learning, remote work, driving, gaming), **mental fatigue and reduced attention span** have become critical problems. Traditional methods to assess cognitive load rely on questionnaires or invasive devices (EEG, eye-trackers), which are **expensive and not user-friendly**.
We need a **non-invasive, low-cost AI solution** that can estimate **cognitive load and attention level** in real-time using **webcam/smartphone cameras**.

---

2. Objective

- To design a real-time system that analyzes **eye movement, blink rate, and pupil dilation** using a normal camera.
- To predict **cognitive load, fatigue, and attention span** using AI/ML models.
- To provide a **dashboard/alerts** for users (e.g., students, drivers, professionals).

---

3. System Features

☑ **Real-time eye tracking** using Mediapipe/OpenCV.
☑ **Blink rate monitoring** (high = fatigue, low = concentration).
☑ **Pupil dilation estimation** (larger pupils = higher cognitive load).
☑ **Gaze tracking** – checks if the user is attentive or distracted.
☑ **Time-series fatigue prediction** using LSTM/GRU.

☑ **Dashboard** to visualize fatigue/cognitive load trends.
☑ **Alerts** (audio/vibration/notification) when attention drops.

---

4. System Modules

1. **Input Acquisition Module**
   - Webcam/phone camera streams user's face & eyes.
2. **Preprocessing Module**
   - Face & eye detection (Mediapipe, Haar Cascade, Dlib).
   - ROI extraction (crop eye region).
3. **Feature Extraction Module**
   - **Blink detection** → Eye aspect ratio (EAR).
   - **Pupil size estimation** → Circular Hough Transform / Thresholding.
   - **Gaze tracking** → Left/right/up/down focus estimation.
   - **Saccade velocity** (eye movement speed).
4. **Cognitive Load Prediction Module**
   - Extract time-series features (blinks/minute, average pupil size, gaze deviation).
   - Feed into ML/Deep Learning model.
   - Output: Cognitive Load Level {Low, Medium, High}.
5. **Alert & Visualization Module**
   - Dashboard showing attention & fatigue trends.
   - Notifications when fatigue detected.

---

5. Workflow (Step by Step)

1. **User opens app** → Camera captures face video.
2. **Face & eye landmarks detected** → Extract eyes & pupil region.
3. **Feature extraction**: Blink rate, pupil size, gaze points.
4. **Features fed into AI model** → Predict cognitive load.
5. **Dashboard updates in real-time** → Graph of attention/fatigue.
6. **If high fatigue detected** → Send alert (beep or push notification).

---

6. Algorithms / Models

- **Face & Eye Detection** → Mediapipe Face Mesh / Dlib 68-landmarks.
- **Blink Detection** → Eye Aspect Ratio (EAR).
- **Pupil Size Detection** → Image threshold + Circular Hough Transform.
- **Gaze Estimation** → Pupil position relative to eye region.
- **Fatigue & Cognitive Load Estimation** →
  - **Machine Learning**: Random Forest, SVM, XGBoost.
  - **Deep Learning**: LSTM/GRU for time-series fatigue patterns.
- **Classification Output** → {Low, Medium, High Cognitive Load}.

7. Expected Output

- **Graph**: Cognitive load vs time.
- **Fatigue alerts**: When user shows reduced attention.
- **Dashboard**: Blink frequency, gaze stability, pupil dilation trend.
- **Accuracy**: Target >85% for cognitive load classification.

---

8. Future Scope

- Extend to **driver drowsiness detection** (integrate with vehicle alert systems).
- Use with **e-learning platforms** to monitor student attention.
- Integrate **EEG or smartwatch data** for multi-modal prediction.
- Develop into **wearable AI glasses** for real-time monitoring.

WORKFLOW :

[Start]

[Camera Capture]

[Detect Face + Eyes]

[Extract Features]

- Blink Rate
- Pupil Dilation
- Gaze Tracking

[Cognitive Load Model]

- LSTM / XGBoost
- Low / Medium / High Load

[Output]

- Dashboard Visualization
- Alert if High Load

[End]